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SUMMARY

Director-level leader (10-15 reports, \$2-4M portfolio) who chose to deepen technical expertise through advanced CS degree and research opportunities while simultaneously serving as sole technical architect for a startup. Now combining executive experience with proven ability to build **enterprise-grade** AI/ML systems from zero to production **in regulated industries**.

EXPERIENCE

Senior Data Engineer & Business Strategist | Colonial Pipeline | Atlanta, Ga | April 2025 - Present

- Enabled advanced analytics across industrial IoT infrastructure by **architecting an intelligent sensor data transformation system** that converts sparse, exception-driven streams into continuous time-series datasets using statistical ML models, automated outlier detection, and business-context-aware flagging algorithms.
- **Led development of a multi-agent AI architecture** for drag-reducing agent injection optimization, orchestrating specialized expert agents (statistical analysis, blending operations expertise, BigQuery integration) with a foundational LLM to migrate legacy spreadsheet analysis into a **scalable web platform** enabling natural language queries, interactive filtering, and automated efficiency optimization recommendations for the power consumption team.
- **Delivered and managed enterprise AI transformation** across multiple business units, **architecting solutions that generated \$1M+ in annual operational savings** through automated vendor selection, optimized fuel blending, and intelligent procurement processes while maintaining strict regulatory compliance.
- **Architected the evolution of a genetic algorithm** from an experimental SQL/notebook fuel blending prototype to a multi-objective optimization engine, designing an extensible **OOP framework** with pluggable fitness functions, selection strategies, and mutation operators, plus automated testing infrastructure for parameter space exploration and performance validation.
- **Orchestrated MLOps pipelines on Vertex AI** with automated CI/CD workflows, custom containerization, and secure deployment practices for regulated industrial environments, enabling scalable ML model lifecycle management.
- **Implemented a comprehensive MLOps framework** using MLflow and Weights & Biases for experiment tracking, model versioning, and performance monitoring, enabling reproducible AI development in security-compliant industrial environments.
- **Designed a dynamic sensor validation system** using SQL workflows that incorporated domain-specific business knowledge into automated data quality metrics, creating **golden tables** that serve as authoritative data sources for downstream analytics.
- **Deployed an intelligent procurement automation system** using Google ADK agents for vendor evaluation and technology stack alignment, implementing automated SWOT analysis and suitability scoring that **delivered \$150K+ annual cost savings** through optimized vendor selection.
- **Built an automated code quality assurance platform** with **RAG architecture** and **vector databases**, implementing event-driven analysis for Dataform SQLX and Python applications to ensure code consistency, security compliance, and continuous improvement.
- **Architected an automated AI model evaluation framework** with intelligent scoring agents and BigQuery analytics, enabling continuous performance monitoring, cost optimization, and data-driven model improvement across production ML systems.

ML/Senior Data Engineer | Gordon Food Services | Advanced Analytics | 2022 - April 2025

- **Transformed an experimental TensorFlow notebook into an enterprise-grade forecasting system**, architecting robust MLOps infrastructure on Vertex AI with automated training pipelines, GPU acceleration, and distributed processing for scalable deployment.
- **Optimized model architecture** using advanced programming and constraint optimization, implementing concurrent and mixed-integer programming, early stopping, and gradient descent to achieve a **4% performance improvement over O9 Solutions' baseline model**.
- **Deployed a production forecasting solution projected to deliver \$1M in annual savings**, validated through comprehensive **five-year historical backtesting** and rigorous performance analysis.
- **Implemented automated model performance monitoring tools**, enabling Data Scientists to leverage Vertex AI Experiments with TensorBoard for demand forecasting and Vizier for streamlined hyperparameter tuning workflows.
- **Engineered end-to-end ML pipelines on GCP**, utilizing Vertex AI for time series forecasting with custom TensorFlow, PyTorch, Prophet, and Greykite models while supporting automated deployment capabilities.
- **Built comprehensive ML infrastructure** including feature selection tools, distributed model training, and support for custom models, BigQuery ML, and AutoML, enabling streamlined deployment and **real-time inference**.
- **Enhanced model training efficiency by 25%** through optimized data access and storage, creating feature stores on BigQuery and leveraging managed datasets and metadata frameworks in GCP.
- **Reduced BigQuery costs by 10% monthly** through strategic partitioning, clustering, and slot usage analysis on core data science tables, demonstrating advanced SQL optimization and cost management expertise.
- **Implemented a comprehensive data governance strategy** using GCP Dataplex with hierarchical metadata tagging and labeling, establishing **data mesh architecture** that enables domain-driven data ownership and discovery across industrial operations.

Research Engineer/Developer | Vanderbilt University | ScopeLab AI Research Lab | 2021 – 2022

- **Architected scalable ETL pipeline infrastructure** using PySpark frameworks, processing **10TB+ of diverse academic and transportation datasets** across 5+ concurrent research projects while establishing automated data preparation workflows that **reduced research scientist setup time from weeks to hours**.
- **Built an enterprise-grade data standardization and ingestion platform** leveraging PySpark SQL and Pandas, creating reusable pipeline templates that **accelerated retraining cycles by 60%** and enabled 5+ researchers to rapidly prototype and deploy solutions.
- **Established comprehensive cloud infrastructure** supporting 5+ research scientists, managing AWS/GCP environments including Jupyter notebooks, compute instances, and distributed processing pipelines with **90%+ uptime SLA** for scalable machine learning experimentation.
- **Developed a full-stack transportation optimization platform** with a **real-time vehicle tracking system** and ML-based route optimization, serving 50+ daily shuttle operations while integrating manual routing adjustments and predictive analytics that **improved operational efficiency by 25%** and reduced passenger wait times.

Technical Lead & Founding Engineer | Mobius.ai | (Transportation optimization SaaS) | 2019 – 2021

- **Sole technical architect** responsible for end-to-end platform development, from initial architecture through production deployment.
- **Built complete data infrastructure** processing **10TB+ datasets**, an ML operations platform, and full-stack applications serving 50+ daily users.
- **Owned the entire technical lifecycle:** architecture design, implementation, model integration, user testing, and iterative improvements based on customer feedback.
- **Established scalable foundations** that enabled 5+ researchers to rapidly prototype and deploy ML solutions.
- The company remains profitable and has operated successfully for over five years.

Director, Business Technology Solutions | Grant Thornton | 2014 – 2019

- **Led a team of 10-15 engineers and consultants**, managing a **\$2-4M annual project portfolio** across Fortune 500 clients.
- **Scaled from Manager to Director in 3 years**, growing practice revenue by **15%** while maintaining **80% team retention**.
- **Delivered \$100M+ in client value** through digital transformation initiatives surrounding tax and financial strategies with large-scale commercial real estate, technology companies, and automated outsourcing solutions.
- **Drove digital transformation initiatives** that generated **multi-million-dollar client savings** by automating financial calculations, aggregating data, and optimizing tax positioning strategies while demonstrating measurable ROI from technology investments.
- **Led large-scale data engineering and migration projects**, standardizing millions of financial records from legacy ERP systems into structured, governed databases while transforming fragmented data into unified, actionable intelligence platforms.
- **Established technical architecture and governance frameworks** to ensure GAAP/IFRS compliance and regulatory adherence, providing domain expertise and maintaining strict data integrity and audit trail requirements across financial systems.
- **Built enduring client relationships** from startups to large corporations, translating technical capabilities into business value through financial analysis, strategic planning, and technology roadmaps that optimized enterprise value and project NPV.

Manager, National Leader Tax Technology Solutions | Cherry Bekaert, LLP | 2012 – 2014

- **Supported practice growth by architecting client acquisition processes**, contributing to doubling practice size and **generating \$2M+ annual revenue** across 30+ concurrent engagements.
- **Led technical hiring decisions and project management** across distributed teams in 5 locations, **improving project success rates by 30%** and **reducing client onboarding time by 40%**.

Senior Associate, Tax Technology & Integration Services | EY, LLP | 2007 – 2011

- **Architected an enterprise data warehousing solution** for the **\$12B MGM CityCenter mega-project**, leading the technical design and implementation of comprehensive analytics infrastructure for **real-time project analytics**.
- **Orchestrated a complex cloud migration** of 500GB+ critical operational data from legacy systems to MicroStrategy Cloud, enabling **self-service analytics** for 100+ enterprise users in the early cloud adoption era.
- Consistently achieved **top-tier performance recognition** (5/5 ratings, top 15% firmwide), demonstrating technical leadership excellence and strategic value delivery in high-stakes enterprise technology implementations.

EDUCATION

Master of Science, Computer Science | Vanderbilt University

Master of Business Administration | Georgia State University

Bachelor of Science, Environmental Engineering | Georgia Institute of Technology

Bootcamp | MongoDB, Express, React, Node.js | Georgia Institute of Technology

CERTIFICATIONS

AWS Certified Developer Associate | 2019-2021

SKILLS

ML/AI:

Python, PyTorch, TensorFlow, scikit-learn, Feature engineering, Prophet, Greykite, AutoML, BQML, TimesFM, OpenAI API, Hugging Face, Prompt Engineering, Generative AI, LangGraph, RAG architectures, Time series forecasting, Anomaly detection

Data Engineering:

BigQuery, SQL, PySpark, Dask, Pandas, NumPy, MySQL, PostgreSQL, Cloud Storage, BigTable, Parquet, JSON, Vertex AI Feature Store, Vertex AI Vector Search, Embeddings, Metadata management

Cloud & MLOps:

GCP, Vertex AI, Google ADK, AWS, Azure, MLflow, Weights & Biases, Airflow, Docker, Kubernetes, CI/CD for ML, Distributed model training, Cloud Run, Vertex AI Pipelines, Dataplex

Development:

JavaScript, Java, React, Node.js, Express, Flask, FastAPI, Custom Container Orchestration, JFrog Artifactory, Snyk

Data Architecture & Governance:

Data mesh architecture, Data governance, Golden table design, Dataplex, Data cataloging, Metadata management, Domain-driven data architecture

Analytics & Visualization:

Looker, Dash Plotly, TensorBoard, Power BI

I build high-performing teams through direct communication, pragmatic problem-solving, and creating environments where people feel genuinely valued and can bring their authentic selves to solve complex technical challenges.
